



OLA END-TO-END DATA ANALYSIS PROJECT

A Complete Data Analytics Project Report

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Tools Used: MySQL | Microsoft Power BI | Microsoft Excel

Dataset Source: Kaggle

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1. Aim of the Project

The aim of this project is to perform a complete end-to-end data analysis on OLA ride booking data to extract meaningful business insights, identify operational patterns, and support data-driven decision making using SQL, Microsoft Power BI, and Microsoft Excel.

2. Objectives

- To extract, clean, and transform raw OLA ride booking data for analysis.
- To write and execute SQL queries to answer key business problem statements.
- To build interactive Power BI dashboards for data visualization and reporting.
- To analyze ride cancellation patterns, booking status, and revenue distribution.
- To evaluate customer and driver ratings across different vehicle types.
- To identify top performing customers, vehicle types, and payment methods.
- To generate actionable business insights to improve OLA operational efficiency.
- To present findings in a structured and professional format for stakeholders.

3. Tools and Technologies Used

- **MySQL** — Used for data extraction, querying, and solving SQL problem statements.
- **Microsoft Power BI** — Used for building interactive dashboards and data visualization.
- **Microsoft Excel** — Used for data cleaning, transformation, and analysis.
- **Kaggle** — Source of the OLA ride booking dataset (Bookings.csv).
- **GitHub** — Used for version control and project hosting.

4. Dataset Information

- **Dataset Name:** Bookings.csv
- **Source:** Kaggle
- **Total Records:** 20,000+ ride booking records
- **Key Columns:** Booking ID, Customer ID, Vehicle Type, Ride Distance, Booking Status, Payment Method, Driver Rating, Customer Rating, Cancellation Reason

5. SQL Problem Statements and Solutions

The following SQL queries were written and executed in MySQL to extract insights from the OLA booking dataset:

Problem Statement 1: Retrieve All Successful Bookings

Description: Retrieve all ride records where the booking was completed successfully.

SQL Query:

```
SELECT * FROM Bookings  
Where Booking_Status = 'Success';
```

Answer / Insight: This query returns all successful ride bookings from the dataset. It helps identify the total number of completed rides and analyze patterns in successful bookings.

Problem Statement 2: Find Average Ride Distance for Each Vehicle Type

Description: Calculate the average ride distance covered by each type of vehicle.

SQL Query:

```
SELECT Vehicle_Type, AVG(Ride_Distance)  
as avg_distance From Bookings  
Group By Vehicle_Type;
```

Answer / Insight: This query groups bookings by vehicle type and calculates the average ride distance for each. It helps identify which vehicle types are used for longer trips.

Problem Statement 3: Get Total Number of Cancelled Rides by Customers

Description: Find the total count of rides that were cancelled by customers.

SQL Query:

Select count(*) from Bookings

Where Booking_Status = 'Canceled by Customer';

Answer / Insight: This query counts all rides cancelled by customers. The result helps understand customer cancellation behavior and identify potential service gaps.

Problem Statement 4: List Top 5 Customers with Highest Number of Rides

Description: Identify the top 5 most active customers based on total number of bookings.

SQL Query:

Select Customer_ID, COUNT(Booking_ID) as Total_rides

FROM Bookings

group by Customer_ID

Order by Total_rides DESC LIMIT 5;

Answer / Insight: This query ranks customers by total bookings and returns the top 5. This insight is valuable for customer retention strategies and loyalty programs.

Problem Statement 5: Get Rides Cancelled by Drivers Due to Personal and Car-Related Issues

Description: Count the number of rides cancelled by drivers specifically due to personal or car-related problems.

SQL Query:

Select count(*) from Bookings

Where Canceled_Rides_by_Driver= 'Personal & Car related issue';

Answer / Insight: This query isolates driver cancellations caused by personal or vehicle issues. It helps management understand supply-side problems affecting ride availability.

5. Power Bi Problem Statements and Solutions

Problem Statements

1. Ride Volume Over Time
2. Booking Status Breakdown
3. Top 5 Vehicle Types by Ride Distance
4. Average Customer Ratings by Vehicle Type
5. cancelled Rides Reasons
6. Revenue by Payment Method
7. Top 5 Customers by Total Booking Value

8. Ride Distance Distribution Per Day
9. Driver Ratings Distribution
10. Customer vs. Driver Ratings

Solutions

Segregation of the views:

1. Overall

- Ride Volume Over Time
- Booking Status Breakdown

2. Vehicle Type

- Top 5 Vehicle Types by Ride Distance

3. Revenue

- Revenue by Payment Method
- Top 5 Customers by Total Booking Value
- Ride Distance Distribution Per Day

4. Cancellation

- Cancelled Rides Reasons (Customer)
- cancelled Rides Reasons(Drivers)

5. Ratings

- Driver Ratings
- Customer Ratings

Answers:

- 1. Ride Volume Over Time:** A time-series chart showing the number of rides per day/week.
- 2. Booking Status Breakdown:** A pie or doughnut chart displaying the proportion of different booking statuses (success, cancelled by the customer, cancelled by the driver, etc.).
- 3. Top 5 Vehicle Types by Ride Distance:** A bar chart ranking vehicle types based on the total distance covered.
- 4. Average Customer Ratings by Vehicle Type:** A column chart showing the average customer ratings for different vehicle types.
- 5. cancelled Rides Reasons:** A bar chart that highlights the common reasons for ride cancellations by customers and drivers.
- 6. Revenue by Payment Method:** A stacked bar chart displaying total revenue based on payment methods (Cash, UPI, Credit Card, etc.).
- 7. Top 5 Customers by Total Booking Value:** A leaderboard visual listing customers who have spent the most on bookings.
- 8. Ride Distance Distribution Per Day:** A histogram or scatter plot showing the distribution of ride distances for different Dates.

- 9. Driver Rating Distribution:** A box plot visualizing the spread of driver ratings for different vehicle types.
- 10. Customer vs. Driver Ratings:** A scatter plot comparing customer and driver ratings for each completed ride, analyzing correlations.

6. Conclusion

This end-to-end OLA Data Analysis project successfully demonstrated the complete data analytics workflow — from raw data extraction and cleaning to SQL querying and interactive Power BI dashboard creation. The project generated valuable business insights related to booking patterns, cancellation behavior, revenue distribution, and customer satisfaction. These insights can directly support OLA management in making data-driven decisions to improve operational efficiency and customer experience

7. Author Information

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